

Property $(\ddagger)$ of the Dual Forces Reflexivity

Automatic math-discovery run `fa_banach_001`

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Source Problem

The source is Deepak Gothwal and T. S. S. R. K. Rao, *Some Geometric Aspects Related to Lim's Condition*, arXiv:2504.09464, first posted in 2025 and revised in 2026 [1]. On page 11 of the current arXiv PDF, immediately before Theorem 18, the authors write:

“We do not know in general if X^* has $(\ddagger)$, then X is reflexive.”

They then prove the affirmative answer under the additional assumption that X contains no isomorphic copy of ℓ_1 .

as an element of $Y^\perp$. Similarly, for each α , ϕ_α is norm-attaining as an element of $Y^\perp$.

Now, $(\ddagger)$ is hereditary. So, $Y^\perp$ has $(\ddagger)$. Thus, (X/Y) has $(\ddagger)$. $\square$

An interesting observation relates reflexivity and Lim's condition in a dual space. If X^* has Lim's condition, then weak* and weak topologies coincide on $S(X^{**})$. By Goldstein's theorem (see [10]), under the canonical embedding, $B(X)$ is w^* -dense in $B(X^{**})$. So, we see that Lim's condition in a dual space implies reflexivity. We do not know in general if X^* has $(\ddagger)$, then X is reflexive. We have a partial result when X has no isomorphic copy of ℓ_1 ($\ell_1 \not\hookrightarrow X$).

Theorem 18. *If X is such that $\ell_1 \not\hookrightarrow X$ and X^* has property $(\ddagger)$, then X is reflexive.*

Proof. Let Y be a separable subspace of X such that $\ell_1 \not\hookrightarrow Y$. By a result in [6, Pg. 215], every element of $S(Y^{**})$ is a w^* -sequential limit of elements of $S(Y)$. Let $\phi \in S(Y^{**})$ be a norm-attaining functional and $\{y_n\}$ be a sequence in $S(Y)$ such that $y_n \xrightarrow{w^*} \phi$. If $\{y_n\}$ does not converge to ϕ in the norm, then $\{y_n\}$ is not Cauchy. Then there is a subsequence, again denoted by $\{y_n\}$ such that for some $\varepsilon > 0$, this subsequence is ε -separated, i.e., $\|y_n - y_m\| \geq \varepsilon$ for all $n \neq m$. Clearly y_n 's are norm-attaining on Y^* and $y_n \rightarrow \phi$ in the w^* -topology.

Now, $Y^\perp \subset X^*$ is a w^* -closed and hence, a proximal subspace. Thus Y^* (being isometrically isomorphic to $X^*/Y^\perp$) has property $(\ddagger)$. So, $\|\phi\| \leq 1 - \varepsilon/2$, a contradiction. Thus, we get that $\phi \in Y$. Now by the Bishop-Phelps theorem, we conclude that Y is reflexive and hence, as reflexivity is a separably determined property, X is a reflexive space. $\square$

Result Type

This packet gives a **candidate full affirmative solution**. It removes the source theorem's extra hypothesis $\ell_1 \not\hookrightarrow X$.

Definition

A Banach space E has property $(\dagger)$ if, whenever $0 < \varepsilon < 2$ and (f_α) is a net of norm-attaining elements of B_{E^*} such that

$$f_\alpha \xrightarrow{w^*} f, \quad \|f_\alpha - f_\beta\| \geq \varepsilon \quad (\alpha \neq \beta),$$

where $f \in B_{E^*}$ is also norm-attaining, then $\|f\| \leq 1 - \varepsilon/2$.

Here an element of E^* is norm-attaining if it attains its norm on B_E . In the source question $E = X^*$, so the relevant net lies in $B_{X^{**}}$ and the weak-star topology is $\sigma(X^{**}, X^*)$.

Proof Intuition

The source paper settles spaces without ℓ_1 , so the missing case is when ℓ_1 embeds into X . James's distortion theorem then supplies a normalized sequence which is a 0.9-copy of the ℓ_1 basis. Its Stone–Čech ultrafilter limits form a weak-star compact family in $B_{X^{**}}$, any two of whose points are at distance at least 1.8.

Those ultrafilter limits need not themselves be norm-attaining. This is the main obstruction, and simply replacing them by Bishop–Phelps approximants would generally destroy convergence. The key device is to first translate the entire separated family so that one chosen limit becomes a nearby norm-attaining functional. Each translated point is then perturbed by a norm-attaining functional with error tending to zero along the net. The perturbed net still converges to the chosen norm-attaining limit and retains a uniform separation. A common rescaling puts everything in the unit ball, where property $(\dagger)$ gives a numerical contradiction.

Three Lemmas

Lemma 1 (Quotient inheritance). *Let E have property $(\dagger)$, and let $Z \subset E$ be a closed proximal subspace. Then E/Z has property $(\dagger)$.*

Proof. Identify $(E/Z)^*$ isometrically with $Z^\perp \subset E^*$. The weak-star topology induced by E/Z agrees on $Z^\perp$ with the restriction of $\sigma(E^*, E)$, because a functional in $Z^\perp$ depends only on the coset of its argument.

It remains to check norm attainment. Suppose $f \in Z^\perp$ attains its norm as a functional on E/Z , say at $x + Z \in S_{E/Z}$. Proximality provides $z \in Z$ such that $\|x + z\| = \|x + Z\| = 1$. Since $f(z) = 0$,

$$|f(x + z)| = |f(x)| = \|f\|.$$

Thus f also attains its norm as an element of E^* . Consequently any net and limit occurring in the definition for E/Z also satisfy the definition's hypotheses in E^* , with the same norms and separation. Property $(\dagger)$ for E gives the required estimate. $\square$

Lemma 2 (An injective net in $\mathbb{N}^*$). *There are a point $p \in \mathbb{N}^* := \beta\mathbb{N} \setminus \mathbb{N}$ and a net of pairwise distinct points $(q_d)_{d \in D} \subset \mathbb{N}^* \setminus \{p\}$ such that $q_d \rightarrow p$. The directed set may be chosen with a map $n : D \rightarrow \mathbb{N}$ satisfying $n(d) \rightarrow \infty$.*

Proof. Write $\mathfrak{c} = 2^{\aleph_0}$. The compact zero-dimensional space $\mathbb{N}^*$ has weight at most $\mathfrak{c}$, while every nonempty open subset of $\mathbb{N}^*$ has cardinality $2^{\mathfrak{c}}$. Indeed, such an open set contains $A^* = \overline{A}^{\beta\mathbb{N}} \setminus A$ for some infinite $A \subset \mathbb{N}$, and A^* has cardinality $2^{\mathfrak{c}}$.

Fix $p \in \mathbb{N}^*$, and choose a neighborhood base $\mathcal{B}$ at p , of cardinality at most $\mathfrak{c}$, closed under finite intersections. Put $D = \mathcal{B} \times \mathbb{N}$, ordered by

$$(U, n) \preceq (V, m) \iff V \subset U \text{ and } m \geq n.$$

Enumerate D and recursively choose $q_{(U,n)} \in U \setminus \{p\}$, never reusing a previously chosen point. At each stage fewer than or equal to $\mathfrak{c}$ points have been used, whereas U has cardinality $2^{\mathfrak{c}}$, so the recursion succeeds. The resulting net is injective, converges to p , and its second coordinate tends to infinity. $\square$

Lemma 3 (The ℓ_1 obstruction). *If a Banach space X contains an isomorphic copy of ℓ_1 , then X^* does not have property $(\dagger)$.*

Proof. By James's distortion theorem [2, 5], there is a sequence $(x_n) \subset S_X$ such that, for every finitely supported scalar sequence (a_n) ,

$$\frac{9}{10} \sum_n |a_n| \leq \left\| \sum_n a_n x_n \right\| \leq \sum_n |a_n|. \quad (1)$$

The map $n \mapsto x_n$, considered in $(B_{X^{**}}, w^*)$, extends continuously to $\Phi : \beta\mathbb{N} \rightarrow (B_{X^{**}}, w^*)$. Put $z_p = \Phi(p)$.

We first record the metric geometry of this compact family. Given distinct $p, q \in \beta\mathbb{N}$, choose $h = (h_n) \in B_{\ell_\infty}$ whose continuous extension to $\beta\mathbb{N}$ has values 1 at p and -1 at q . On $\text{span}\{x_n\}$, set

$$f_h \left(\sum_n a_n x_n \right) = \frac{9}{10} \sum_n a_n h_n.$$

The lower estimate in (1) gives $\|f_h\| \leq 1$, so Hahn–Banach extends f_h to an element of B_{X^*} . Therefore

$$\|z_p - z_q\| \geq |(z_p - z_q)(f_h)| = \frac{9}{5}. \quad (2)$$

Taking $h_n = 1$ also gives

$$\frac{9}{10} \leq \|z_p\| \leq 1. \quad (3)$$

Fix $p \in \mathbb{N}^*$. Apply the Bishop–Phelps theorem to the Banach space X^* [4]: choose a norm-attaining $w \in X^{**}$ with

$$\|w - z_p\| < \frac{1}{10}. \quad (4)$$

Let $(q_d)_{d \in D}$ and $n(d)$ be supplied by Lemma 2. Write $a = w - z_p$. For every d , Bishop–Phelps again gives a norm-attaining $v_d \in X^{**}$ such that

$$\|v_d - (z_{q_d} + a)\| < \frac{1}{10n(d)}. \quad (5)$$

Since $q_d \rightarrow p$, we have $z_{q_d} + a \xrightarrow{w^*} w$; the errors in (5) tend to zero in norm, hence $v_d \xrightarrow{w^*} w$.

Set

$$\phi_d = \frac{5}{6} v_d, \quad \phi = \frac{5}{6} w.$$

Both the ϕ_d 's and ϕ are norm-attaining. Moreover, (4) and (5) show that

$$\|v_d\| < 1 + \frac{1}{10} + \frac{1}{10} = \frac{6}{5},$$

so $\phi_d \in B_{X^{**}}$; also $\phi \in B_{X^{**}}$, and $\phi_d \xrightarrow{w^*} \phi$. For $d \neq e$, (2) and (5) yield

$$\|\phi_d - \phi_e\| > \frac{5}{6} \left(\frac{9}{5} - \frac{1}{10} - \frac{1}{10} \right) = \frac{4}{3}.$$

On the other hand, (3) and (4) give

$$\|\phi\| = \frac{5}{6} \|w\| > \frac{5}{6} \left(\frac{9}{10} - \frac{1}{10} \right) = \frac{2}{3}.$$

If X^* had property $(\ddagger)$, applying it with $\varepsilon = 4/3$ would instead imply $\|\phi\| \leq 1 - (4/3)/2 = 1/3$, a contradiction. $\square$

Main Theorem

Theorem 1. *For every Banach space X , if X^* has property $(\ddagger)$, then X is reflexive.*

Proof. Suppose first that $\ell_1 \hookrightarrow X$. Lemma 3 says that X^* fails property $(\ddagger)$, contrary to the hypothesis.

It remains to treat $\ell_1 \not\hookrightarrow X$. We include the source paper's argument, making the quotient step explicit. If X were nonreflexive, separable determination of reflexivity would give a separable nonreflexive closed subspace $Y \subset X$. Then $\ell_1 \not\hookrightarrow Y$. The annihilator $Y^\perp$ is proximal in X^* : if $x^* \in X^*$, extend $x^*|_Y$ to $g \in X^*$ without increasing its norm; then $x^* - g \in Y^\perp$ and $\|g\| = \text{dist}(x^*, Y^\perp)$. Since $Y^* \cong X^*/Y^\perp$, Lemma 1 shows that Y^* has property $(\ddagger)$.

The Odell–Rosenthal theorem [3] says that every element of $B_{Y^{**}}$ is a weak-star sequential limit of a bounded sequence in Y . Let $u \in S_{Y^{**}}$ be norm-attaining. We may choose a sequence $(y_n) \subset S_Y$ with $y_n \xrightarrow{w^*} u$: start with the bounded sequence from Odell–Rosenthal and normalize it, using a norming functional for u to see that its norms tend to one. If $u \notin Y$, this sequence cannot be norm Cauchy, and a subsequence is ε -separated for some $\varepsilon > 0$. Every y_n , viewed in Y^{**} , is norm-attaining on Y^* , as is u . Property $(\ddagger)$ for Y^* would give $1 = \|u\| \leq 1 - \varepsilon/2$, impossible. Hence every norm-attaining element of Y^{**} belongs to Y .

By Bishop–Phelps, norm-attaining elements of $Y^{**} = (Y^*)^*$ are norm dense in Y^{**} . Since the canonical copy of Y is norm closed, it follows that $Y = Y^{**}$, contradicting the choice of Y . Thus X is reflexive. $\square$

Corollary 1. *The open question on page 11 of Gothwal–Rao has an affirmative answer, with no additional structural hypothesis on X .*

Verification Notes

The proof was audited along the following fault lines.

- The Stone–Čech map is used only into the weak-star compact Hausdorff ball $B_{X^{**}}$, so its continuous extension exists by the universal property of $\beta\mathbb{N}$.
- Pairwise separation is obtained from Hahn–Banach extensions whose norm bound follows in the correct direction from the lower ℓ_1 estimate.
- The net in Lemma 2 is genuinely injective; a sequence would not suffice because $\mathbb{N}^*$ has no nontrivial convergent sequences.
- Bishop–Phelps is first applied to the desired limit, then to translated points with errors tending to zero. This is what preserves both a norm-attaining limit and weak-star convergence.
- The common factor $5/6$ places the whole net in the unit ball. The separation lower bound is $4/3$, while the limit norm is strictly greater than $2/3$; property $(\ddagger)$ would force it to be at most $1/3$.
- Quotient inheritance is proved directly and does not require a separate heredity assertion for arbitrary subspaces.

The reusable script `code/check_constants.py` verifies the numerical margins, including small perturbations of the three chosen tolerances. It is only an arithmetic check; the proof is analytic and topological.

Deep-Upgrade Audit

An initial partial route used an asymptotically isometric copy of ℓ_1 , for which the ultrafilter points themselves share an exact norming face. That hypothesis is not automatic: Dowling–Johnson–Lennard–Turett [5] construct equivalent norms on ℓ_1 containing no asymptotically isometric copy. This invalidates the tempting upgrade from James’s theorem. The translated Bishop–Phelps construction above is the additional deep attempt that removes the false step and works with the fixed-error almost-isometric copy actually guaranteed by James.

Novelty and Literature Check

The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2504.09464, the exact title, “Lim’s condition”, “property ($\dagger$)”, “dual”, and “reflexive”. No duplicate result or attempt was found. The current arXiv source (revised February 2026) still states the general problem as open and proves only the no- ℓ_1 case.

A bounded external search on August 11, 2026 used the exact paper title, arXiv id, the quoted open sentence, the authors’ names, and combinations of “property ($\dagger$)”, “dual”, and “reflexive”. It found the source record and generic mirrors, but no later paper or independent theorem stated as resolving this question. The proof does use classical results of James, Odell–Rosenthal, and Bishop–Phelps; the particular translation-and-vanishing- perturbation combination answering the source question was not found. Accordingly, novelty confidence is moderate rather than definitive.

Limitations and Human Review Recommendation

This is a candidate full solution subject to expert review, not a claim of established publication priority. The recommended review focus is Lemma 2’s cardinal selection and the two-stage Bishop–Phelps construction in Lemma 3. If those steps are accepted, the constants give a strict contradiction and the source theorem covers the complementary no- ℓ_1 case.

References

- [1] D. Gothwal and T. S. S. R. K. Rao, *Some Geometric Aspects Related to Lim’s Condition*, arXiv:2504.09464, 2025; revised 2026.
- [2] R. C. James, *Uniformly non-square Banach spaces*, Ann. of Math. (2) **80** (1964), 542–550.
- [3] E. Odell and H. P. Rosenthal, *A double-dual characterization of separable Banach spaces containing ℓ_1* , Israel J. Math. **20** (1975), 375–384.
- [4] E. Bishop and R. R. Phelps, *A proof that every Banach space is subreflexive*, Bull. Amer. Math. Soc. **67** (1961), 97–98.
- [5] P. N. Dowling, W. B. Johnson, C. J. Lennard, and B. Turett, *The optimality of James’s distortion theorems*, Proc. Amer. Math. Soc. **125** (1997), 167–174.